

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 0.0004985120715100572
rotor delay [s]: 0.153009804634
gimbal delay [s]: 0.034852580194
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 4.269539040625534e-17
common-scale FD null norm: 1.498728700551765e-11

Dimensionless coordinate

log_mass_scale	-0.0848175201997
log_second_moment_mode_xx	0.741257148678
log_second_moment_mode_yy	0.9734268173
log_second_moment_mode_zz	-0.580899062268
log_second_moment_mode_xy	-0.03924252449
log_second_moment_mode_xz	1.45077650819
log_second_moment_mode_yz	0.0389329505569
dimensionless_cog_displacement_x	-0.181214837572
dimensionless_cog_displacement_y	0.00804646573566
dimensionless_cog_displacement_z	-0.286167007668
log_force_effectiveness_1	-0.119033585838
log_force_effectiveness_2	-0.375907181814
log_force_effectiveness_3	-0.400164950255
log_force_effectiveness_4	-0.153861665605

Physical representative

mass [kg]: 2.160365462221414
CoG body [m]: [-0.03674275 0.00151106 -0.04531555]
force effectiveness: [0.88777798 0.68666607 0.67020949 0.85739062]
inertia [kg m²]:
[0.1713802049, 0.003255371102, -0.007784727093]
[0.003255371102, 0.1876940733, -0.0001090435878]
[-0.007784727093, -0.0001090435878, 0.3580376615]
principal moments [kg m²]: [0.1704445 0.18830532 0.35836212]

Per-bag raw residual-wrench RMS

failure_2_single: [0.17530699 0.28292575 0.48693993 0.04751981 0.12897259 0.05910763]